```python
# Online Python compiler (interpreter)
# Write and run Python online using this editor.
m = [
"MONAR",
"CHYBD",
"EFGIK",
"LPQST",
"UVWXZ"
]

p = input("Enter plaintext: ")
c = ""

if len(p) % 2 != 0:
    p += "X"

for i in range(0, len(p), 2):

    a = p[i]
    b = p[i+1]

    for r in range(5):
```

```
Enter plaintext: INSTRUMENTS
Ciphertext: GATLMZCLRQXA
```

main.py

Visualize

Run ▶

```
    for r in range(5):
        if a in m[r]:
            r1 = r
            c1 = m[r].index(a)

        if b in m[r]:
            r2 = r
            c2 = m[r].index(b)

    if r1 == r2:
        c += m[r1][(c1+1)%5]
        c += m[r2][(c2+1)%5]

    elif c1 == c2:
        c += m[(r1+1)%5][c1]
        c += m[(r2+1)%5][c2]

    else:
        c += m[r1][c2]
        c += m[r2][c1]

print("Ciphertext:", c)
```

Output

```
Enter plaintext: INSTRUMENTS
Ciphertext: GATLMZCLRQXA
```